

DATASET FOR THE QUESTIONS (a)–(g)

ITA0613 – Machine Learning

Dataset: FAO_CROP_DATASET

Dataset creator / uploader: Sujal Dhanwani

Geographic scope: India, district level

Time period: 2020–2024

Size: 18,000 records and 12 columns

Record unit: District-year observation

License stated on the dataset page: Apache 2.0

Dataset Purpose

The dataset provides district-level annual information combining climate conditions, soil-related indicators and crop-yield information. It is used as the common dataset for Questions (a) through (g) in the machine-learning assignment.

Main Variables Used in the Assignment

Variable group	Variables / role
Time & location	Year, State, District
Agriculture	Crop / crop-yield information
Climate	Rainfall, Temperature, Humidity
Soil	Soil nutrient / soil-health indicators
Target	Crop Yield

Use Across Questions

- (a) Cleaning, missing-value handling and feature engineering.
- (b) From-scratch k-NN regression with Euclidean and Mahalanobis distances.
- (c) From-scratch locally weighted regression and bias-variance comparison.
- (d) Discretised candidate-elimination / version-space analysis.
- (e) Scaling, runtime/memory analysis and approximate-search prototype.
- (f) Climate-yield visualisations and policy interpretation.
- (g) Uncertainty, fairness and SDG 2 / SDG 13 discussion.

Source and Credit

Kaggle dataset: **FAO_CROP_DATASET**, uploaded by **Sujal Dhanwani**. The public dataset page describes it as an India district-level climate, soil and crop-yield dataset covering 2020–2024, with 18,000 rows and 12 columns and an Apache 2.0 license.

Source: <https://www.kaggle.com/datasets/sujaldhanwani/fao-crop-dataset>